Yun Zhang

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EDUCATION

Johns Hopkins University (Expected Graduation: 12/2026) Baltimore MD, United States
M.S.E. in Electrical and Computer Engineering 01/2025 - Present

Massachusetts Institute of Technology – IDSS Remote
Data Science and Machine Learning: Making Data-Driven Decisions 04/2024-09/2024

University College Dublin & Beijing University of Technology Beijing, China
Dual Bachelor's degrees in Software Engineering 09/2020-07/2024

- GPA: 3.59/4.2 (Second Class Honors, Grade 1)   Ranking: top 10%

Technical Skills

Programming Languages: Python, C++, MATLAB, SQL, Java
Machine Learning / Data Science: TensorFlow, scikit-learn, XGBoost, NumPy, pandas, Matplotlib
Research Areas: Python for Data Science; Descriptive/Inferential Statistics; Spectral Clustering; Regression&Prediction; Classification&Hypothesis Testing; Deep Learning; Networking&Graphical Models Reinforcement Learning, Machine Learning, Deep Learning

FULL-TIME JOB

Deloitte Shenzhen, China
Analyst – IT & Specialized Assurance – Risk Advisory 09/2024 – 12/2024

- Conducted IT audits for three publicly listed companies, assessing IT security, data governance, and compliance while evaluating database infrastructure, firewall security, and data protection strategies.
- Advanced Data Analytics & Risk Identification: Leveraged Deloitte's internal data analytics tools, Microsoft Excel (advanced functions, macros), Power BI, and SQL to clean, transform, and analyze large datasets for IT risk assessments, anomaly detection, and internal control enhancements.
- Regulatory & Compliance Oversight: Provided key insights on internal control weaknesses, cybersecurity threats, and IT governance improvements, ensuring compliance with SOX, GDPR, and other industry regulations.
- High-Pressure Stakeholder Communication: Among the only two newly hired IT Specialist Risk Advisor in Shenzhen, worked directly with senior executives, IT directors, and regulatory bodies, delivering audit findings, risk assessments, and strategic IT control recommendations in high-stakes environments.

INTERNSHIP

Siemens Energy Beijing, China
Management Assistant 07/2023-11/2023

- Assisted the CIO (Chief Information Officer) and IT Department in Siemens Energy Hub CN, supporting management activities, IT daily operations, projects, etc.
- Organized project meetings and business events; managed cost claiming and end-user support; provided technical support to test, trial run and acceptance of business applications
- Planned the charity activities between Siemens Energy and Baimiao Zi Central School of Tumert Left Banner, Inner Mongolia. As the lecturer, provided artificial intelligence science courses to students, and provided technical support for artificial intelligence car training during the activities.

SAS Institute Beijing, China
Data Analyst 04/2023-07/2023

- Conduct data research and technical background study for the digital transformation of banking industry

Project

Interpreting Pain Scales in Knee X-rays (Johns Hopkins University - BME) 01/2025 – 05/2025

- Analyzed 6,705 right-knee samples from the NIH Osteoarthritis Initiative (OAI) baseline cohort (>4,000 participants).
- Built XGBoost regression models predicting continuous KOOS/WOMAC pain scores from 32 radiographic features and demographic covariates.
- Applied TreeSHAP for global feature attribution and implemented SANE (Sufficient & Necessary Explanations) via brute-force combinatorial search over 37 predictors.

- ✧ Conducted fairness diagnostics using residual gap analysis and propensity-score matching to quantify racial pain disparities after structural adjustment.
- ✧ Demonstrated increase in explained variance (R^2 from 0.042 to 0.092) when integrating demographic variables.

Surgical Segmentation & Safety Alert System (JHU PULSE Lab) 01/2025 – 04/2026

Photoacoustic & Ultrasound Systems Engineering Lab (Director: Muyinatu A. Lediju Bell, Ph.D.)

- ✧ Processed cadaver surgical datasets collected in collaboration with Johns Hopkins Hospital.
- ✧ Implemented ROI cropping, field-of-view normalization, and pixel-level mask refinement.
- ✧ Developed monocular geometric distance estimation module to compute tool-to-ureter proximity and classify safety levels (Safe / Caution / Danger).
- ✧ Integrated computer vision modeling with safety-aware decision pipeline in high-stakes surgical simulation context.

Deep Operator Learning for Spatio-Temporal Workload Prediction (JHU-CE) 01/2026 – 05/2026

✧ Modeled workload dynamics in space–air–ground computing systems using a diffusion–reaction PDE and developed a numerical simulation pipeline to generate synthetic spatio-temporal data.

✧ Implemented MLP, DeepONet, FNO, and physics-informed FNO models to learn the workload evolution operator and predict future workload fields.

Algorithms for Sensor-Based Robotics (Johns Hopkins University – CS) 09/2025 – 12/2025

- ✧ Implemented an Expansive Space Tree (EST) sampling-based motion planner in C++ using ROS2 and MoveIt!, including nearest-neighbor search, configuration weighting, collision checking, and bidirectional tree expansion for UR5 path planning.

Predicting Potential Customers (MIT IDSS Project) 07/2024 - 09/2024

- ✧ Developed ML models (Logistic Regression, Decision Trees) to classify customer data, optimizing sales team focus on high-value prospects and boosting conversion rates.
- ✧ Performed EDA, data preprocessing (missing value imputation, one-hot encoding, normalization), and feature selection (correlation analysis, tree-based methods) to reduce model complexity.
- ✧ Evaluated models using accuracy, precision, recall, and F1-score; delivered actionable insights to refine sales/marketing strategies.

FoodHub Order Analysis (MIT IDSS Project) 06/2024-07/2024

- ✧ Analyzed restaurant demand trends to improve business strategies and customer experience.
- ✧ Applied data visualization (Matplotlib, Seaborn) and preprocessing (missing value handling, normalization) to identify key features and generate strategic recommendations.
- ✧ Translated insights into actionable business opportunities for operational optimization.

AI Medical Personalized Health Platform (Undergraduate Graduation Project) 02/2024 - 03/2024

- ✧ Developed a personalized platform for users to monitor physical and mental health, providing guidance to improve users' health with assistant by OpenAI API and original smart algorithms
- ✧ Responsible for data processing, quality verification in the group. Designed, developed and maintained database, optimized database performance, ensuring data consistency and security; designed ‘ body health’ module recommendation algorithms in the role of back-end developer

Machine Learning Data-Science Job-Change Prediction 09/2023 -11/2023

- ✧ Built ML models (Naïve Bayes, KNN, Random Forest) to predict candidates’ post-training employment intentions. Performed comprehensive data preprocessing (missing value handling, outlier detection, dimensionality reduction) and visualization using Python (Pandas, NumPy, Matplotlib).
- ✧ Optimized model selection via cross-validation and hyperparameter tuning, achieving high precision in predictions.

Shopping Website Design & Development 09/2022 - 12/2022

- ✧ Independently built SQLite databases, all back-end functions under the flask framework
- ✧ Completed part of the front-end web framework design and data transfer front and back, and finished most of the HTML files and part of the front and back-end AJAX connections for the website

EXTRACURRICULAR ACTIVITY

College Squad Soccer Team *Co-Founder* 10/2020 - 03/2024

- ✧ Co-founded the first women's soccer team of Beijing University of Technology, and participated in the Beijing University women's soccer competition with the school team every year

AWARD

- ✧ Beijing-Dublin College Scholarship & BJUT Scholarship 2021-2022/22-23/23-24